Feng Yu

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Education

Ph.D in Computer Science, University of Exeter, UK. Sep 2023 – Present
Research interests: Federated Learning, Continual Learning, Foundation Models, and Efficient AI.
Supervisor: *Prof. Jia Hu* and *Prof. Geyong Min*.

M.Eng in Cyberspace Security, Fujian Normal University, China. Sep 2020 – Jun 2023
Research interests: Federated Learning, Edge Computing, and Privacy Protection. GPA: 87.88/100
Supervisor: Prof. *Hui Lin* and Prof. *Xiaoding Wang*.

B.Eng in Computer Science and Technology, Hunan University of Science and Technology, China. Sep 2016 – Jun 2020
Thesis: Application of recommendation algorithm based on RL (**Outstanding**). GPA: 3.34/4.0
Advisor: Prof. *Qiongbin Zhang*.

Publications

[1] **Feng Yu**, Hui Lin, Xiaoding Wang, Sahil Garg, Georges Kaddoum, Satinder Singh, and Mohammad Mehedi Hassan. "Communication-Efficient Personalized Federated Meta-Learning in Edge Networks". In: *IEEE Transactions on Network and Service Management* 20.2 (June 2023), pp. 1558–1571. DOI: 10.1109/TNSM.2023.3263831.

[2] **Feng Yu**, Hui Lin, Xiaoding Wang, Abdussalam Yassine, and M. Shamim Hossain. "Blockchain-Empowered Secure Federated Learning System: Architecture and Applications". In: *Computer Communications* 196 (Dec. 2022), pp. 55–65. DOI: 10.1016/j.comcom.2022.09.008.

[3] Jianmin Liu, Xiaoding Wang, Hui Lin, and **Feng Yu**. "GSAA: A Novel Graph Spatiotemporal Attention Algorithm for Smart City Traffic Prediction". In: *ACM Transactions on Sensor Networks* (Nov. 2023). DOI: 10.1145/3631608.

[4] Ankita Sharma, Shalli Rani, Syed Hassan Shah, Rohit Sharma, **Feng Yu**, and Mohammad Mehedi Hassan. "An Efficient Hybrid Deep Learning Model for Denial of Service Detection in Cyber Physical Systems". In: *IEEE Transactions on Network Science and Engineering* 10.5 (Sept. 2023), pp. 2419–2428. DOI: 10.1109/TNSE.2023.3273301.

Preprint

[1] **Feng Yu**, Jia Hu, and Geyong Min. "Efficient Federated Class-Incremental Learning of Pre-Trained Models via Task-agnostic Low-rank Residual Adaptation". In: *arXiv preprint arXiv:2505.12318* (2025).

[2] Zi Wang, Fei Wu, **Feng Yu**, Yurui Zhou, Jia Hu, and Geyong Min. "Federated continual learning for edge-ai: A comprehensive survey". In: *arXiv preprint arXiv:2411.13740* (2024).

Experience

Research Assistant, University of Exeter Jan 2025 - Apr 2025
Involved: Matlab, UI design
- Design and develop the user interface of the Smart Shipping Acceleration Fund Digital Twin.

R & D Intern (ISG), Lenovo Computer Technology Co. LTD, Shanghai Jun 2022 – Sep 2022
Involved: Shell/Java/Python/Web programming, BUG fix, Log platform Design
- Responsible for ESXI VM and LXCA installation automation script programming; Familiar with Git, Maven, Jenkins, shell and PowerShell script syntax;
- Learn the BMC, IPMI, Redfish, LXCA platforms, and the use of Jira and Bugzilla; Master LXCA CFC project back-end code workflow, solve several bugs in CFC project;
- Build CFC Server prompt functionality with the Dojo framework; Lean the programming and development of Server Configuration Pattern function module;
- Design and create the **log platform** based on ELK and Kafka cluster (3 nodes), and further optimize Kafka performance.

Undergraduate Graduation Thesis, HNUST Dec 2019 – Jun 2020
Involved: Python, Reinforcement Learning, Recommendation algorithm, Bandit algorithm design
- Develop the collaborative algorithms via matrix factorization and bandit algorithms;
- Apply Python to extract preference features from movie data and propose a novel recommendation algorithm based on the idea of gambling machines.

Service account development, HNUST, Team leader Jun 2018 – May 2019
Involved: Java/Web programming, System Design, Software Engineering

- Participate in the back-end development of the **College's Alumni Association service account**;
- Responsible for the back-end development, front-end and back-end test docking and service launch of the **Waku Yiren pilot mutual aid service account**.

Text Intelligence Challenge, Dagan Data, Team member — Jul 2018 – Oct 2018

Involved: Python, SVM, TF-IDF, XGBoost, Machine Learning and Ensemble Learning

- Build a model to predict the category of text through the body of long text data;
- Apply machine learning algorithms (such as SVM, TF-IDF, XGBoost, etc.), and obtain a **Top 10% score**.

ACM-ICPC training team, HNUST, Team member — Dec 2016 – Jun 2018

Involved: Data Structure, Algorithm, C++, Java and Python programming

- Master **basic data structure**, such as stack, queue, linked list, binary tree, graph, and lookup set, tree array, line tree, bisect graph, etc.
- Master **basic algorithms**, such as dichotomy, DFS, BFS, dynamic programming, shortest path algorithm, etc.

Service and Teaching

Reviewer

- **Conference:** IEEE ICDCS'24, TrustCom'24, IEEE IUCC'24
- **Journal:** IEEE IoTJ

Conference Organisation

- IEEE HPCC'25, *Web and System Management Chair* — Aug 2025
- IEEE TrustCom'23, *Online Session Chair* — Nov 2023
- IEEE TrustCom'23, *Reception Volunteer* — Nov 2023

Teaching Assistant (TA)

- ECM 2428 IT Project Management, *PTA*, UNEXE — Sep 2024 - Dec 2024
- ECM 2414 Software Development, *PTA*, UNEXE — Sep 2024 - Dec 2024
- ECM 2433 The C Family, *PTA*, UNEXE — Jan 2024 – Apr 2024
- Computer Networks, *TA*, FJNU — Sep 2020 – Jan 2021
- Programming Experiment (Data structure and C/C++ programming), *TA*, HNUST — 2017 – 2019
- Object Oriented Programming, *TA*, HNUST — 2017 – 2018
- Advanced Mathematics and Professional Basic Courses, *CTA*, HNUST — 2017 – 2018

Main Honors and Awards

China Scholarship Council and University of Exeter **(*CSC-Exeter*) Scholarships** — 2023

Second-class Academic Scholarship in FJNU — 2022

Outstanding Graduate in Hunan Province and HNUST (**Top** ***3%***) — 2020

Outstanding Undergraduate Graduation Thesis of College in HNUST — 2020

National Encouragement Scholarship (twice), **Outstanding Student Leader** in HNUST — 2017 & 2019

First prize (twice) in HNUST — 2017 & 2019

Merit Student (twice), College Merit Student Model in HNUST — 2018 & 2019

Second prize, **Merit Student** (twice), College Merit Student Model, in HNUST — 2017 – 2019

Competences

Languages Mandarin (*native*), English (*working proficiency*, IELTS 6.5)

Programming Languages: C/C++, Python, Golang, Java, SQL, HTML, CSS, JavaScript, Bash, Matlab.

Frameworks & Tools: Git, LaTeX, PyTorch, TensorFlow, FATE, FedLab, Flower, FISCO BCOS, Numpy, Pandas, Scikit-learn, CUDA, Flask, ElasticSearch, Logstash, Kibana, Kafka, SpringBoot, React, Angular, Docker, Redis.

Certificates: Intermediate Software Designer, CCF Certified Software Professional Top 10% (Sep 2018).

Interests

Sports: Running, Cycling, Marathon, Climbing, Triathlon, Badminton, Table Tennis, Basketball, Extreme Sport.

Misc: Travelling, History, Writing and Learn new technologies, such as federated learning, continual learning, foundation models, edge intelligence, deep reinforcement learning, generative adversarial network, few-shot learning, recommendation system and related applications.

Last Updated: *May 2025*